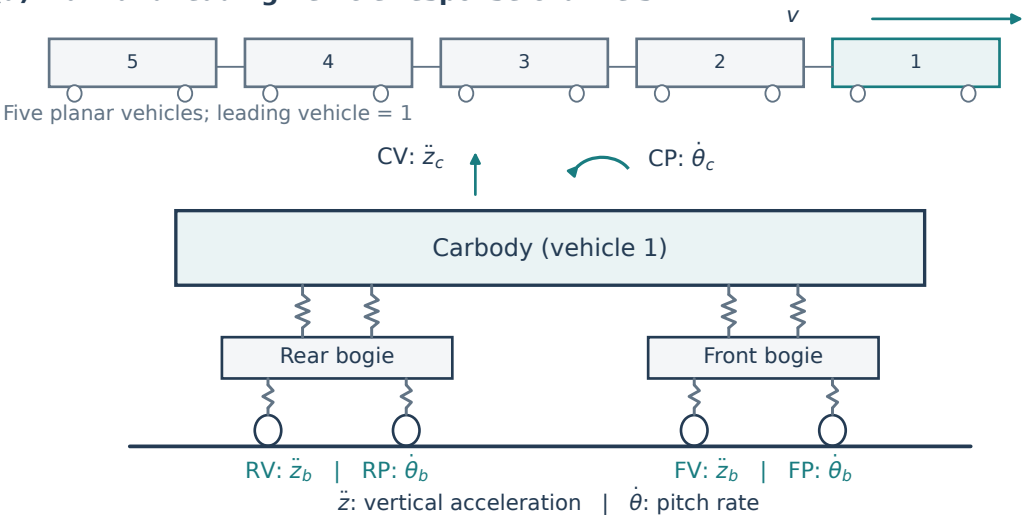
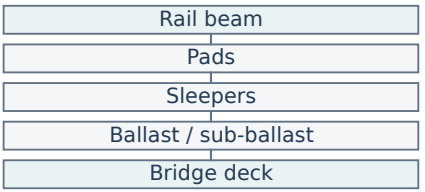
(a) Train and leading-vehicle response channels

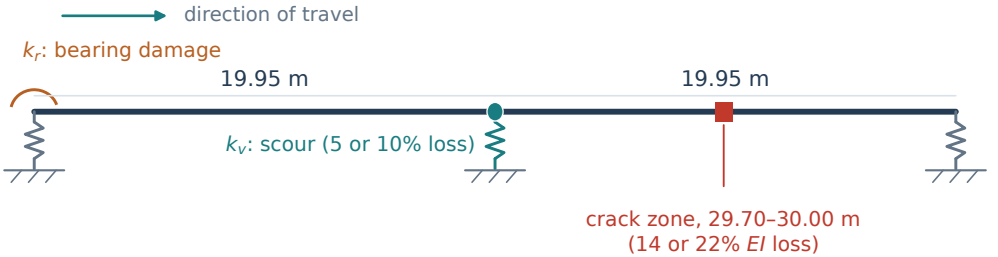


Wheelsets follow the rail (no independent DOFs); their accelerations are not used

(b) Track model



Rail: beam on discrete supports, one per 0.60 m sleeper bay.

On the bridge, the ballast mass is lumped at the deck nodes; there is no shear coupling between ballast blocks.

(c) Bridge and damage locations